The convinced atheist objection

The hypothesis that an intelligent designer exists is really, really unlikely in the first place, so premise (3) is false.

the fine-tuning argument

1. $Pr(e \mid \text{design}) = 0.5$
2. $Pr(e \mid \text{non-design}) = \dfrac{1}{10^{120}}$
3. P(design)=P(non-design)=0.5.
4. Bayes' theorem.
5. P(design) ≈ 1 (1,2,3)
6. If the universe was created by an intelligent designer, then God exists.

———————————————-

C. P(God exists) ≈ 1.

Skepticism about physics

The physics is wrong (or at least might be).

The anthropic objection

We could never observe the falsity of the claim that the constants permits life since, if it were false, we would not exist to observe it.

Let’s look at one last objection:

The multiverse objection

There are very many — perhaps infinitely many — distinct universes, which can have different fundamental physical constants.

Surely, this objection goes, if there were enough universes, then no matter how improbable it is that one of them would have constants that fall in the life-permitting range, it is not so improbable that some would.

Our situation would then be somewhat analogous to the position of someone who wins the lottery. The odds of **that** person winning the lottery were very small; but if enough people buy tickets, it is not so improbable that someone wins.

It would be unreasonable for the winner to infer that the lottery was rigged in her favor; just so, it would be unreasonable for us to assume that our universe was designed.

If we take seriously the multiverse hypothesis, that again causes us to revise our view of the probability of life given non-design.

fine-
tuning of the
universe

Bayes' theorem

the argument

objections

Suppose that we think that there is a 0.5 probability that we live in a multiverse, and think that the probability of life existing in a multiverse is 1 (or very close to it). Then (setting aside for now the possibility that contemporary physics is mistaken) that makes the probability of life given non-design 0.5. And in that case life is equally likely on the design hypothesis and on the non-design hypothesis. And then then fact of life is **no evidence at all** for the design hypothesis.

So the key question is: do we have good reason to think that the multiverse hypothesis is true?

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follows.

Enough about probability (for now). Let's turn now to some results from contemporary physics which will be important to the argument which

A first question: what does the theory provided by physics include?

“The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.” (Hawthorne & Isaacs, “Fine-tuning fine-tuning”)

These "dimensionless constants" will be our focus.

energy density of empty space.

One such constant is the cosmological constant, which measures the

The cosmological constant is just a number which (as far as we know) could have different values consistent with the laws of physics.

(like the cosmological constant) would fall in a certain range.

One thing that the standard model of physics gives us is a measure of how likely it is, given the laws of the nature, that the fundamental constants hard to see how life could have evolved. If there were no planets, it is hard to see how life could have evolved.

life to have evolved. For example, if there were nothing but hydrogen, it is

We can make certain plausible assumptions about what it would take for

Given assumptions such as these, we can look at what the standard model of physics tells us about how likely it is that, for example, the cosmological constant has a value which would permit the evolution of life.

“Physicists have determined the (approximate) values of the fundamental constants by measurement. (There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

supporting value, given the laws of nature, is very low.

This is the claim that that the cosmological constant having a life-

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. "Small" here doesn't mean "1 in 10,000" or "1 in 1,000,000". It means the kind of fraction that one would resort to exponents to describe, as in "1 in 10 to the 120". The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

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The claim is that, according to current physics, the probability of the cosmological constant falling in a life-supporting range, given the laws of

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fine-
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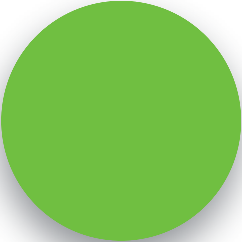
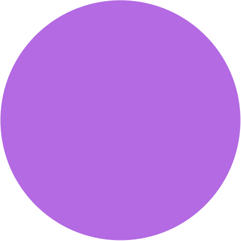
Bayes' theorem





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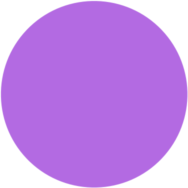
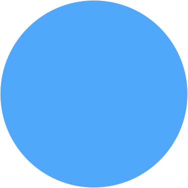
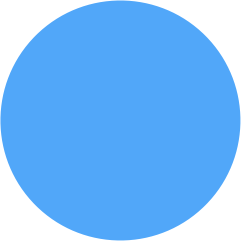




the
argument



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Let's now ask how to turn these facts about current physics and probability theory into an argument for the existence of God.

Much as we considered two different hypotheses about the number of balls in the urn, so we can consider two different hypotheses about the origins of the universe.

Let the **design hypothesis** be the hypothesis that the universe was created by an intelligent designer. Let the **non-design** hypotheses be the hypothesis that it was not.

Let our **evidence** (e) be the fact that the cosmological constant has a value in the life-permitting range.

Then what do we know about the relevant probabilities?

$$Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}$$

$$Pr(e \mid \text{design}) = 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h) * P(e|h)}{P(e)}$$

design hypothesis given our evidence is a matter of just plugging in the

With this information in hand, figuring out the probability of the nonnumbers.

$$Pr(\text{non-design}) = 0.5$$

$$Pr(\text{design}) = 0.5$$

$$Pr(e) = 0.25$$

$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

consider the odds of winning Powerball one trillion times in a row. Call that a “super

It is difficult to think about numbers as large as the denominator of this fraction. But to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now

Powerball."

that a “super duper Powerball.”

Now consider the odds of winning a super Powerball one trillion times in a row. Call

Now consider the odds of winning a super duper Powerball one trillion times in a row.

odds of the universe being life-permitting by chance.

The odds of this happening are about $1 / 10^{44}$ — so much, much higher than the

This means that if you simply take the physics at face value, and begin by assigning a probability of 0.5 to the non-design hypothesis, you should think that the chances of the non-design hypothesis being true are vastly lower than the chances of winning a super duper powerball a trillion times in a row.

minus the very small number we were just discussing:

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1

$$Pr(\mathrm{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.

objections

1. P

2. P

8. If the universe was created by an

7. P(design) $\approx$ 1. (5,6)

4. Bayes' theorem.

5. P

3. P(design)=P(non-design)=0.5.

. (1,2,3,4)

C. $P(\text{God exists}) \approx 1$. (7,8)

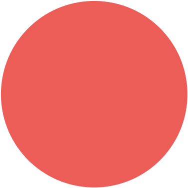
intelligent designer, then God exists.

6. Life exists.

$$(\texttt{life}\,|\,\texttt{design}) = 0.5$$

$$(\texttt{life}|\texttt{non-design}) = \frac{1}{10^{120}}$$

$$(\texttt{design}|\texttt{life}) \approx 1$$



fine-
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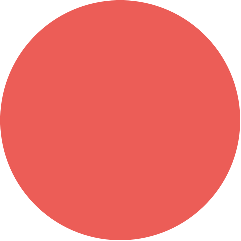

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